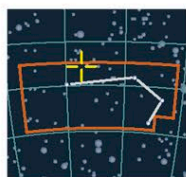


IRREGULAR CLUSTER

Sculptor Group



ALTERNATIVE NAME
South Polar Group

DISTANCE 9 million light-years to center

NUMBER OF GALAXIES
19 (6 major)

BRIGHTEST MEMBER
NGC 253 (8.2)

SCULPTOR

Lying just beyond the gravitational boundaries of the Local Group, the Sculptor Group is similar in size to

the Local Group. It is also a young cluster of irregular and spiral galaxies, with no major ellipticals. It is possible that this group, the Local Group, and another group called Maffei 1 were once part of the same larger cluster.

The closest member to Earth is NGC 55, an irregular galaxy that, like the Large Magellanic Cloud (see p.310), shows enough structure for some astronomers to consider it a single-armed spiral. The dominant galaxy, however, is NGC 253. This large spiral is the same size as the Milky Way and more than twice the size of any other galaxy in the group.

GALAXY NGC 253

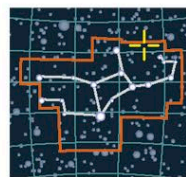
This large spiral dominates the Sculptor Group in this wide-field image. Most of the other galaxies are too faint to be seen without powerful telescopes.

STARBURST GALAXY

NGC 253 is a spiral starburst galaxy—a galaxy undergoing a surge of star formation. The surge may have been triggered by a series of supernovae.

IRREGULAR CLUSTER

Virgo Cluster



ALTERNATIVE NAME
Virgo I Cluster

DISTANCE 52 million light-years to center

NUMBER OF GALAXIES
2,000 (160 major)

BRIGHTEST MEMBER
M49 (9.3)

VIRGO

The Virgo Cluster is the nearest galaxy cluster worthy of the name; it is a dense collection of galaxies at the heart of the larger supercluster to which the Local Group also belongs. The contrast with smaller galaxy

“groups” is striking—the Virgo Cluster contains around 160 major spiral and elliptical galaxies crammed into a volume little larger than that of the Local Group, along with more than 2,000 smaller galaxies. At its heart lie the giant ellipticals M87 (see p.323), M84, and M86, which are thought to have formed from the collisions of spirals over billions of years. Each giant elliptical seems to be at the center of its own subgroup of galaxies—the cluster has not yet settled to become uniform. The cluster’s gravity influences a huge region, extending as far as the Local Group and beyond—the Milky Way and its neighbors are falling toward the Virgo Cluster at 900,000 mph (1.4 million kph).

CENTER OF THE CLUSTER

The Virgo Cluster’s core has a high density of large galaxies. The two bright galaxies on the right are the ellipticals M84 and M86.

EXPLORING SPACE

X-RAY IMAGING AND CLUSTER GAS

Many galaxy clusters are strong sources of X-rays, and orbiting X-ray telescopes can reveal features that remain hidden in visible-light images. While some X-ray sources are located at the centers of the cluster galaxies, the majority of radiation often comes from diffuse gas clouds, independent of the individual galaxies. The process that strips gas out of the cluster galaxies (see p.327) also heats it to generate the X-rays. The distribution of gas offers clues to a cluster’s age and history.

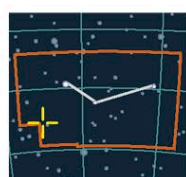


FORNAX IN X-RAYS

This image of the Fornax cluster shows X-ray-emitting gas in blue. Both central galaxies have trailing plumes of gas, suggesting that the entire cluster is moving through sparser clouds.

REGULAR CLUSTER

Fornax Cluster



CATALOG NUMBER
Abell S 373

DISTANCE 65 million light-years to center

NUMBER OF GALAXIES
54 major galaxies

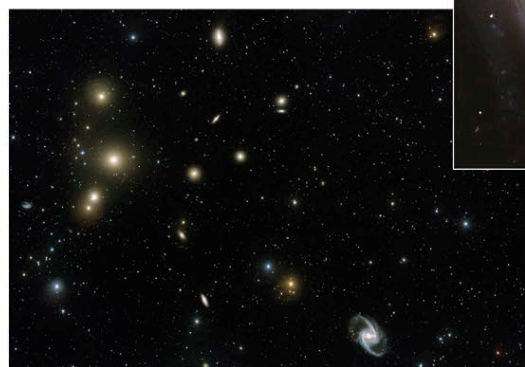
BRIGHTEST MEMBER
NGC 1316 (9.8)

FORNAX

Fornax is home to a relatively nearby galaxy cluster, centered at around the same distance as the Virgo Cluster. However, the Fornax Cluster is at a later stage of evolution than the younger Virgo group. Here, spiral galaxies are rare—the cluster’s major galaxies are

GALAXY NGC 1365

One of the Fornax Cluster’s few spirals, NGC 1365 has a dust bar through its core.



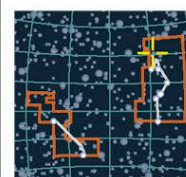
CLUSTER CORE

In the Fornax Cluster’s central region lie NGC 1399 (center left) and NGC 1365 (bottom right). As a rule, elliptical galaxies predominate.

mostly ellipticals, distributed evenly around the giant elliptical NGC 1399. Dwarf galaxies lying between the major ones are also mostly small ellipticals, suggesting that the cluster formed long ago and that interactions between its galaxies have had time to strip away most of their star-forming gas (see p.327). This account of the cluster’s evolution has recently been confirmed by the orbiting Chandra X-ray Observatory (see panel, left).

COMPACT GROUP

Seyfert’s Sextet



CATALOG NUMBERS
NGC 6027 and NGC 6027A–C

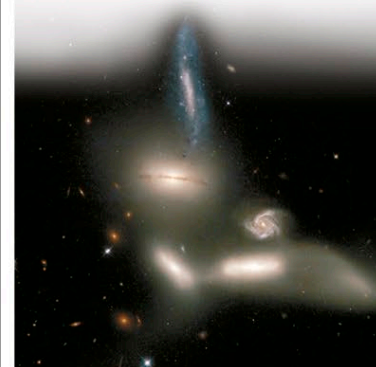
DISTANCE
190 million light-years

NUMBER OF GALAXIES 4

BRIGHTEST MEMBER
NGC 6027 (14.7)

SERPENS

Seyfert’s Sextet actually contains just four members—each a misshapen spiral galaxy locked to the others in a gravitational waltz within a region of space no larger than the Milky Way. The sextet, as seen from Earth, is completed by a small face-on spiral that happens to lie in the background and by a distorted star cloud (at lower right in the image below).



QUARTET PLUS TWO